psymetadata example: stasielowicz2019a

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```r
library(metaselection)
library(metadat)
library(psymetadata)
```

```
## Warning: package 'psymetadata' was built under R version 4.5.3
```

```r
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.2

## here() starts at C:/GitHub/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

Example Meta-analysis

```r
dat_ex <- stasielowicz2019a

dat_ex <- dat_ex |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )
```

Unadjusted estimate: mean only

```r
ex_unadj_mean <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML"
)
ex_unadj_mean
```

```
##
## Random-Effects Model (k = 76; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0321 (SE = 0.0064)
## tau (square root of estimated tau^2 value):      0.1791
## I^2 (total heterogeneity / total variability):   85.87%
## H^2 (total variability / sampling variability):  7.08
##
## Test for Heterogeneity:
## Q(df = 75) = 655.9085, p-val < .0001
##
## Model Results:
##
## estimate      se    zval    pval   ci.lb   ci.ub
##   0.2163  0.0228  9.4820  <.0001  0.1716  0.2610  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

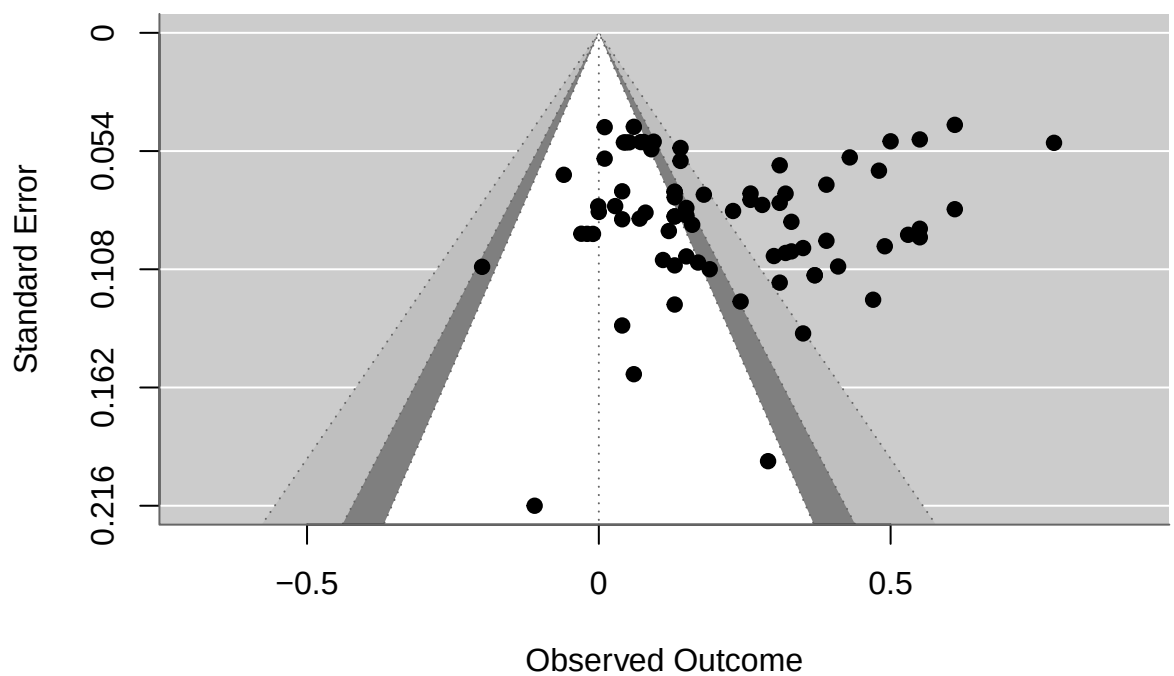
```r
ex_unadj_mean_cil <- ex_unadj_mean$tau2 - 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_ciu <- ex_unadj_mean$tau2 + 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_cil; ex_unadj_mean_ciu
```

```
## [1] 0.0195958
```

```
## [1] 0.04454241
```

Forest plot

```r
funnel(ex_unadj_mean, refline = 0, level = c(90, 95, 99))
```



Unadjusted estimate: moderator

```r
ex_unadj_mod <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML",
```

```r
  mods = ~ adapt_method
)
ex_unadj_mod
```

```
##
## Mixed-Effects Model (k = 76; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.0197 (SE = 0.0043)
## tau (square root of estimated tau^2 value):             0.1404
## I^2 (residual heterogeneity / unaccounted variability): 78.77%
## H^2 (unaccounted variability / sampling variability):   4.71
## R^2 (amount of heterogeneity accounted for):            38.55%
##
## Test for Residual Heterogeneity:
## QE(df = 74) = 359.4168, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 32.9366, p-val < .0001
##
## Model Results:
##
##               estimate      se     zval    pval    ci.lb    ci.ub
## intrcpt         0.5296  0.0577   9.1761  <.0001   0.4165   0.6428  ***
## adapt_method   -0.2176  0.0379  -5.7390  <.0001  -0.2919  -0.1433  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ex_unadj_mod_cil <- ex_unadj_mod$tau2 - 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_ciu <- ex_unadj_mod$tau2 + 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_cil; ex_unadj_mod_ciu
```

```
## [1] 0.01127316
```

```
## [1] 0.0281409
```

Adjusted estimate in metafor: Beta model

```r
ex_adj_mf <- selmodel(
  x = ex_unadj_mod,
  type = "beta",
  steps = c(0.025, 0.975),
  control=list(optimizer="nlminb"),
  #verbose = "TRUE"
)
```

```
## Warning: Error when trying to invert the Hessian.
```

```r
ex_adj_mf
```

```
##
## Mixed-Effects Model (k = 76; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity): 0.0367
## tau (square root of estimated tau^2 value):         0.1915
##
## Test for Residual Heterogeneity:
## LRT(df = 1) = 161.9343, p-val < .0001
##
## Model Results:
##
##               estimate  se  zval  pval  ci.lb  ci.ub
## intrcpt         0.6748  NA    NA    NA     NA     NA
## adapt_method   -0.3913  NA    NA    NA     NA     NA
##
## Test for Selection Model Parameters:
## LRT(df = 2) = 13.9240, p-val = 0.0009
##
## Selection Model Results:
##
##          estimate  se  zval  pval  ci.lb  ci.ub
## delta.1    1.2301  NA    NA    NA     NA     NA
## delta.2    2.4481  NA    NA    NA     NA     NA
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ex_adj_mf_cil <- ex_adj_mf$tau2 - 1.96*ex_adj_mf$se.tau2
ex_adj_mf_ciu <- ex_adj_mf$tau2 + 1.96*ex_adj_mf$se.tau2
ex_adj_mf_cil; ex_adj_mf_ciu
```

```
## [1] NA
```

```
## [1] NA
```

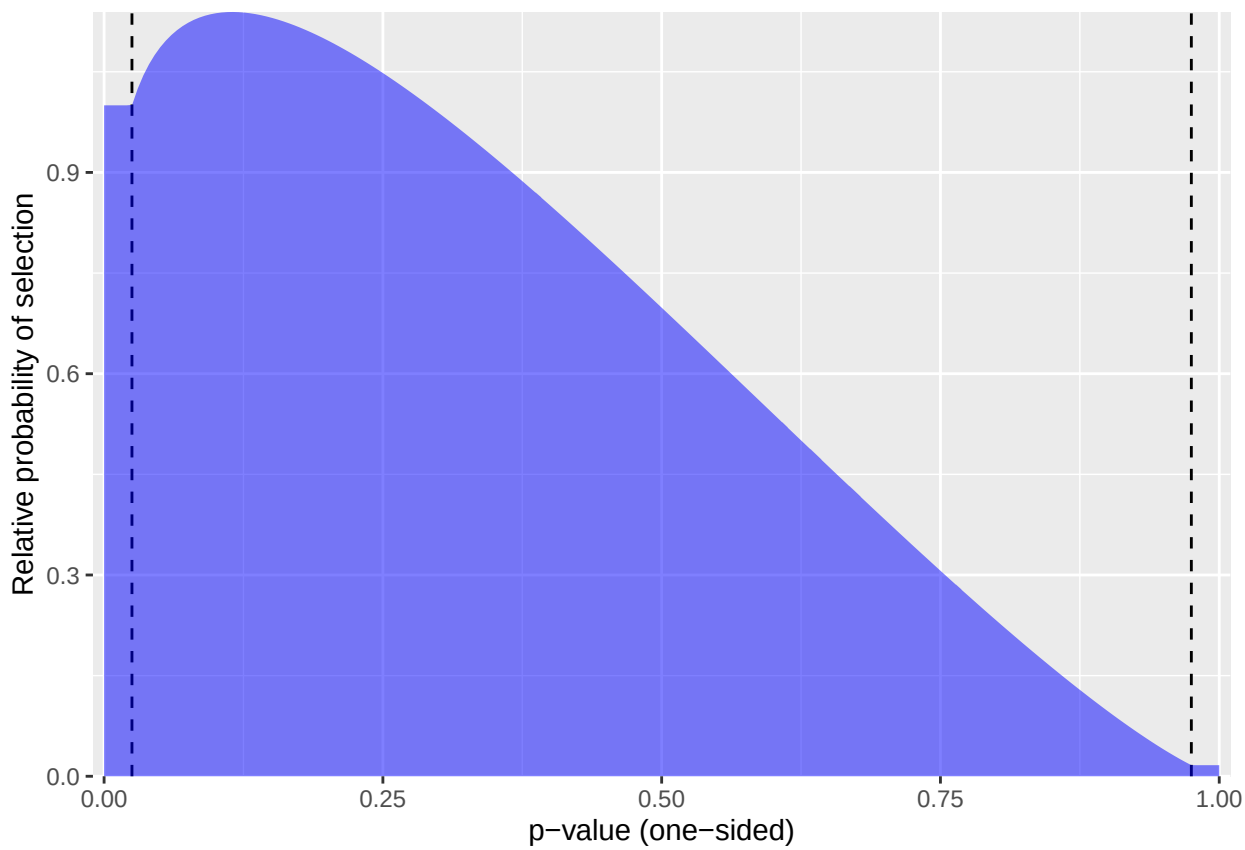
Adjusted estimate in metaselection: Beta model mean only

```r
ex_adj_ms_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)
ex_adj_ms_mean
```

```
##    param    Est     SE  p_value  CI_lo CI_hi
##     beta 0.0346 0.2055 0.866227 -0.368 0.437
##     tau2 0.0763 0.0303       NA  0.035 0.166
##  lambda1 1.1671 0.1783 0.311996  0.865 1.575
##  lambda2 2.2868 0.5731 0.000965  1.399 3.737
```

Selection plot: Beta model mean only

```
selection_plot(ex_adj_ms_mean)
```



Adjusted estimate in metaselection: Beta model moderator

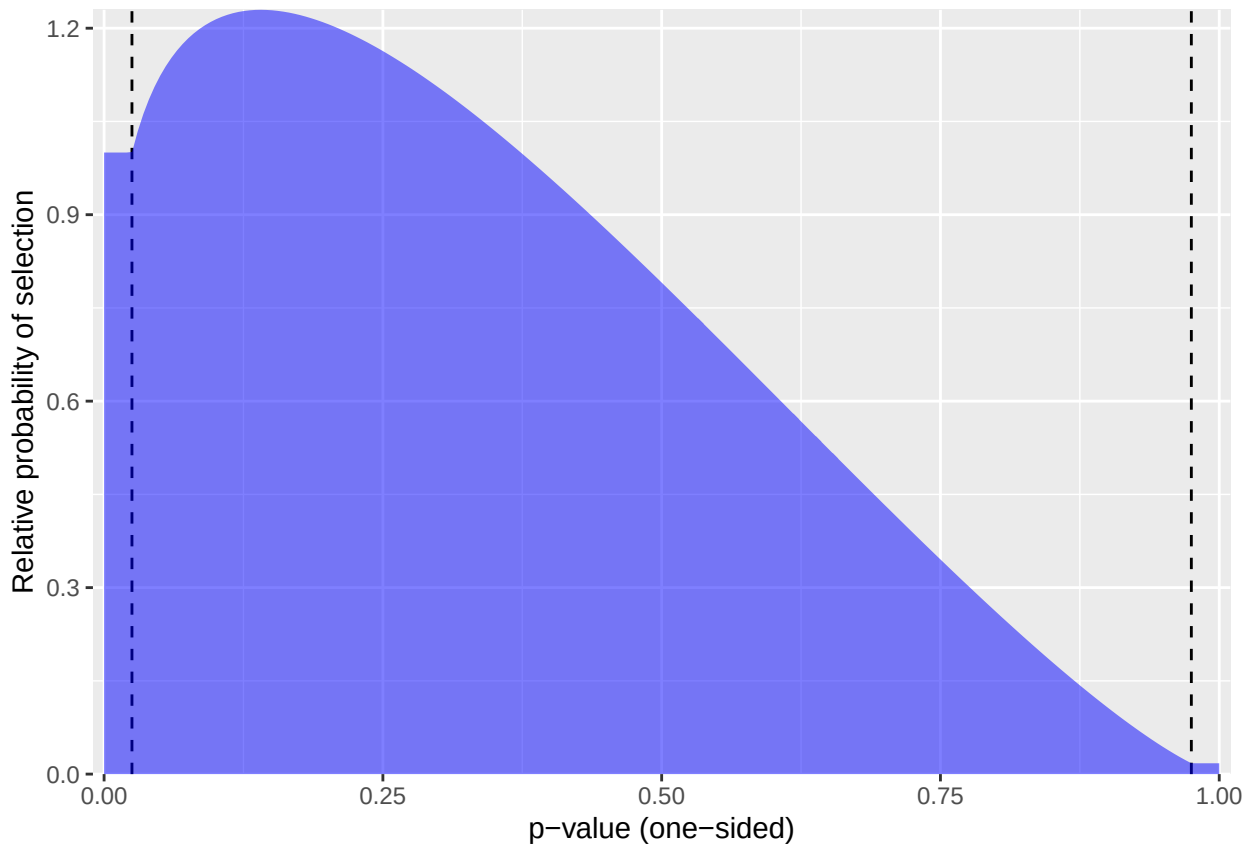
```
ex_adj_ms_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
```

```
  make_sandwich = TRUE,
  mean_mods = ~ adapt_method
)
ex_adj_ms_mod
```

```
##              param     Est    SE  p_value   CI_lo   CI_hi
##   beta_(Intercept)  0.6522 0.139 2.79e-06  0.3794  0.9250
##  beta_adapt_method -0.3666 0.114 1.30e-03 -0.5900 -0.1432
##               tau2  0.0349 0.011       NA  0.0187  0.0648
##            lambda1  1.2163 0.200 2.33e-01  0.8817  1.6778
##            lambda2  2.3217 0.581 7.67e-04  1.4214  3.7923
```

Selection plot: Beta model moderator

```
selection_plot(ex_adj_ms_mod)
```



CHE estimate

```r
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
    obs = esid,
    data = dat_ex,
    rho = 0.5,
    sparse = TRUE
  )

ex_che <- rma.mv(yi = yi,
                 V = V_mat,
                 random = ~ 1 | studyid / esid,
                 data = dat_ex,
                 sparse = TRUE,
                 digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)

ex_che
```

```
##
## Multivariate Meta-Analysis Model (k = 76; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.022  0.147     35     no       studyid
## sigma^2.2  0.011  0.104     76     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 75) = 616.937, p-val < .001
##
## Number of estimates:   76
## Number of clusters:    35
## Estimates per cluster: 1-8 (mean: 2.17, median: 1)
##
## Model Results:
##
## estimate     se¹   tval¹     df¹   pval¹  ci.lb¹  ci.ub¹
##    0.182  0.032   5.760   32.71   <.001   0.118   0.246   ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

Adjusted estimate in metaselection: Step 3PSM model

```r
ex_adj_ms_step3 <- selection_model(
  dat = dat_ex,
```

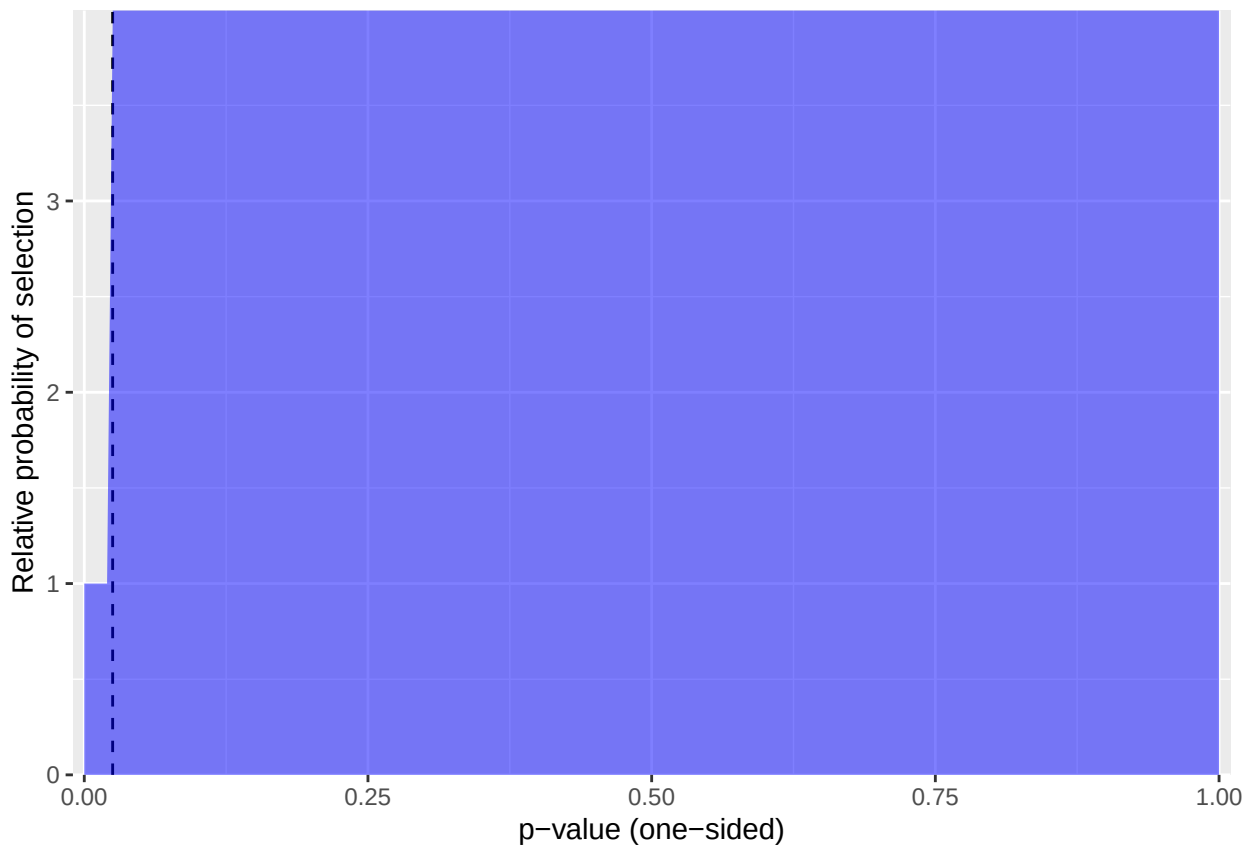
```
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

ex_adj_ms_step3
```

```
##    param    Est      SE  p_value  CI_lo  CI_hi
##     beta 0.3094 0.03926 3.26e-15 0.2324 0.3863
##     tau2 0.0273 0.00642       NA 0.0173 0.0433
##  lambda1 3.9954 1.73657 1.44e-03 1.7045 9.3654
```

Selection plot: Step 3 PSM model

```
selection_plot(ex_adj_ms_step3)
```

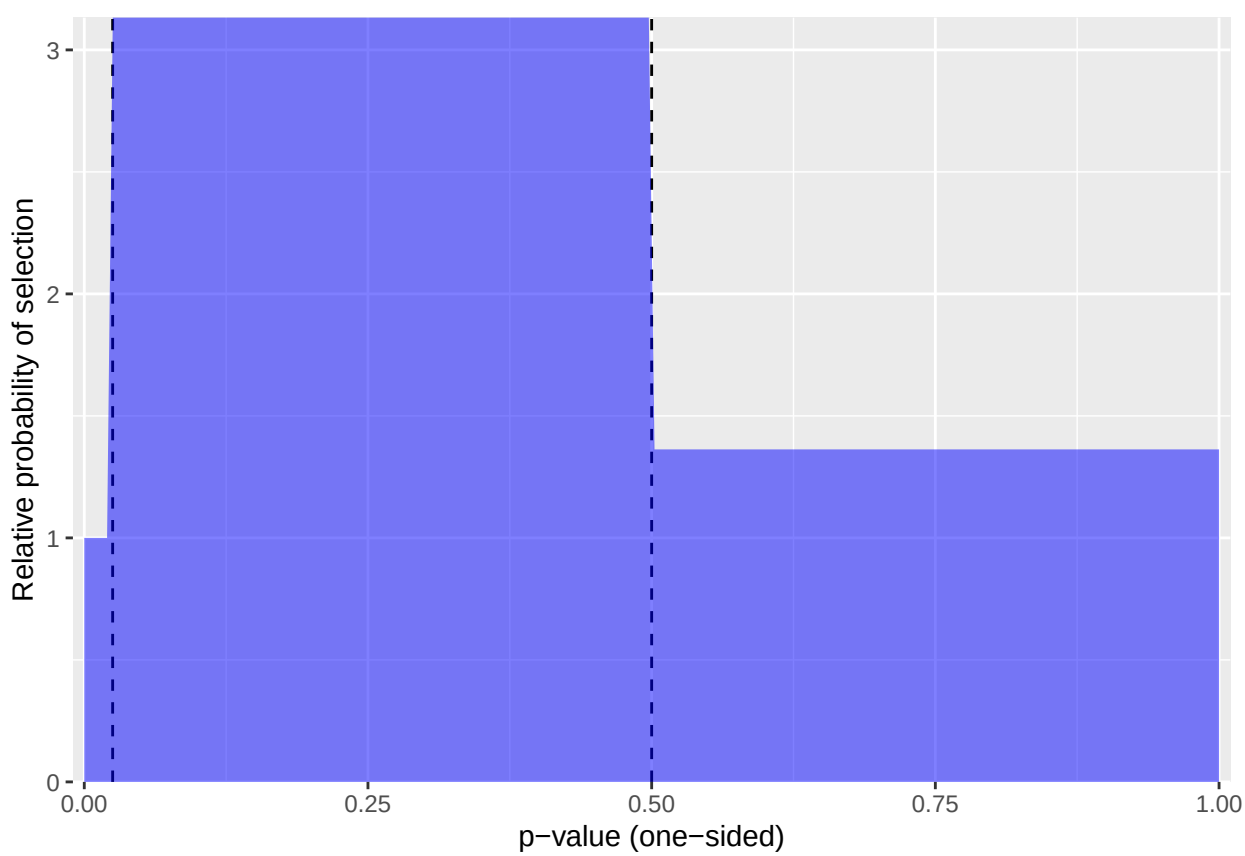


Adjusted estimate in metaselection: Step 4PSM model

```
ex_adj_ms_step4 <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

ex_adj_ms_step4
```

```
##    param   Est      SE  p_value  CI_lo  CI_hi
##     beta 0.274 0.04140 3.47e-11 0.1931 0.3554
##     tau2 0.035 0.00844       NA 0.0218 0.0561
##  lambda1 3.132 1.10120 1.16e-03 1.5727 6.2391
##  lambda2 1.363 0.68381 5.37e-01 0.5097 3.6437
```

Selection plot: Step 4PSM model

```
selection_plot(ex_adj_ms_step4)
```